5. Problem Statement

Organizations frequently store and share sensitive data such as

- Names
- Email Addresses
- Phone Numbers
- Aadhar Numbers
- PAN Numbers
- Addresses
- Any Other Sensitive Information

Manual sanitization is error-prone and inefficient. There is a need for an automated system that:

- Detects PII in multiple data formats
- Removes or masks sensitive information
- Maintains auditability and security

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6. What Is PII?

Personally Identifiable Information (PII) refers to any data that can be used to identify or trace an individual's identity, either directly or when combined with other information.

PII Identification Criteria

- Name: Rakesh Sharma
- Aadhaar Number: 5487 8795 5678
- DOB: 12/05/1978
- Email: XXXXX@gmail.com

- Card Number: 1234 5678 9876 4321
- Name on Card: Anjali Verma
- CVV: 489
- Expiry Date: 12/26

Non PII

Roll No	102345
Aadhar Number	123456789012
Date of Birth	12/05/1997
Card Number	1234567898764321

Note: Non-Personally Identifiable Information (Non-PII) refers to data that cannot be used to identify an individual directly or indirectly. It does not contain names, contact details, or any relational information that could link the data to a specific person.

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7. User roles

Admin (Privileged User)

Capabilities:

- Upload structured and unstructured data
- View original data
- View sanitized output
- Manage users
- View audit logs
- Download PII-free files

Supported upload formats:

- **Structured:** SQL, CSV, JSON
- **Unstructured:** PDF, DOCX, TXT
- **Media:** Images (PNG, JPG)
- Any other structured/Unstructured file.

Non- Admin /Standard User

Capabilities:

- View sanitized data only
- Download PII-free files
- Search and filter records

Restrictions:

- Cannot access the original raw data
- Cannot bypass sanitization

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8. Key Features

Multi-Format Data Handling

The Platform Must Support:
Data Type: Examples
Structured: SQL dumps, CSV, JSON
Unstructured: PDF, DOCX, TXT
Images: ID proofs, screenshots

Processing Pipeline

- Admin uploads file
- File is stored securely
- PII detection engine scans data
- PII is removed or masked
- Sanitized version is stored
- Only sanitized version is accessible to non-admin users

PII Detection Engine

- Names
- Phone numbers
- PAN numbers
- Email addresses
- Aadhaar numbers
- IP addresses
- Addresses

Detection techniques

- Regex-based detection
- NLP-based entity recognition (NER)
- ML models (optional bonus)

Access Control

- Role-Based Access Control (RBAC)
- Admin can view raw + sanitized data
- Non-admin can only view sanitized data

Secure Storage

- Encryption at rest
- Encryption in transit (HTTPS)
- Access controls

Audit Logging

- File uploads
- User access
- Download actions
- PII detection events

PII Removal / Masking

Masking:
[johndoe@email.com] (mailto:johndoe@email.com) → j***@email.com

Redaction: Replace with "[REDACTED]"
Tokenization: Replace with a unique token

Example:
Original: "My name is Rahul Sharma, and my Aadhaar is 1234-5678-9012"
Sanitized: "My name is [REDACTED], and my Aadhaar is [REDACTED]"

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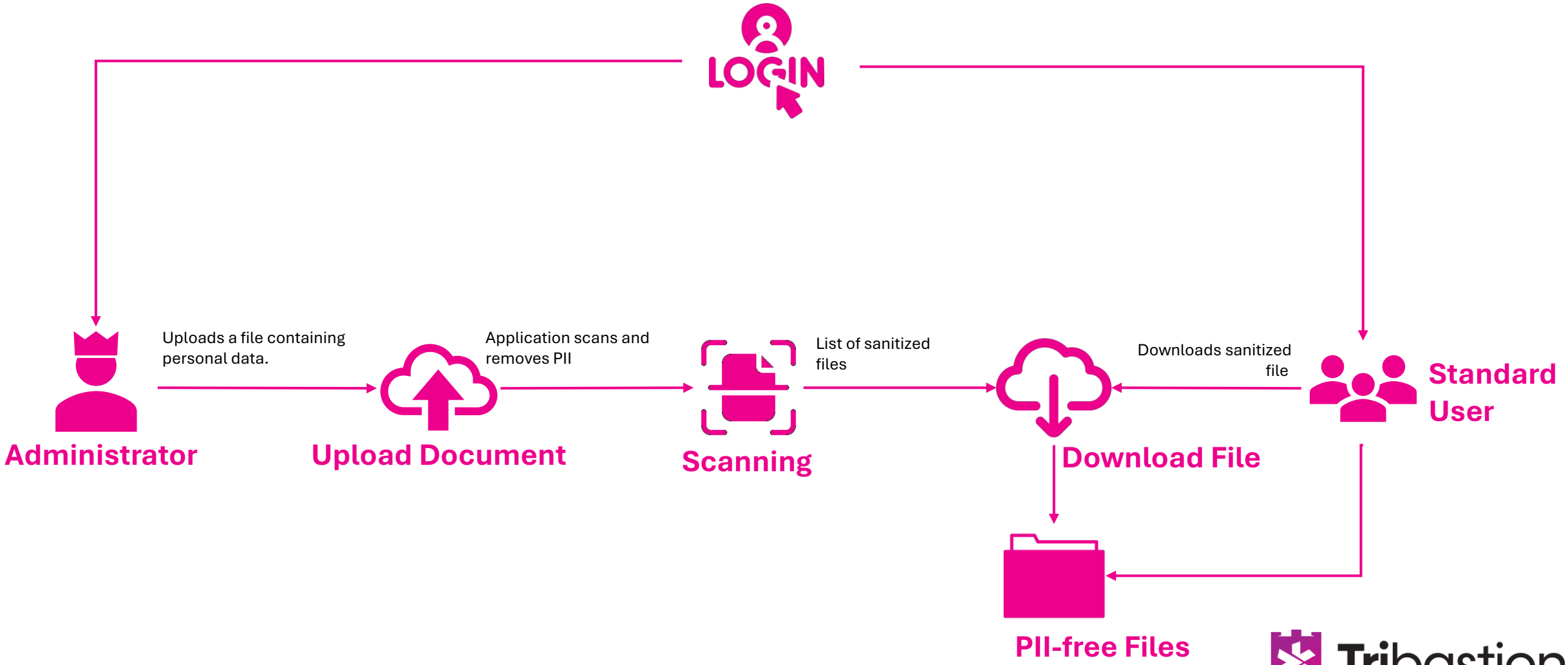
9. Sample Workflow



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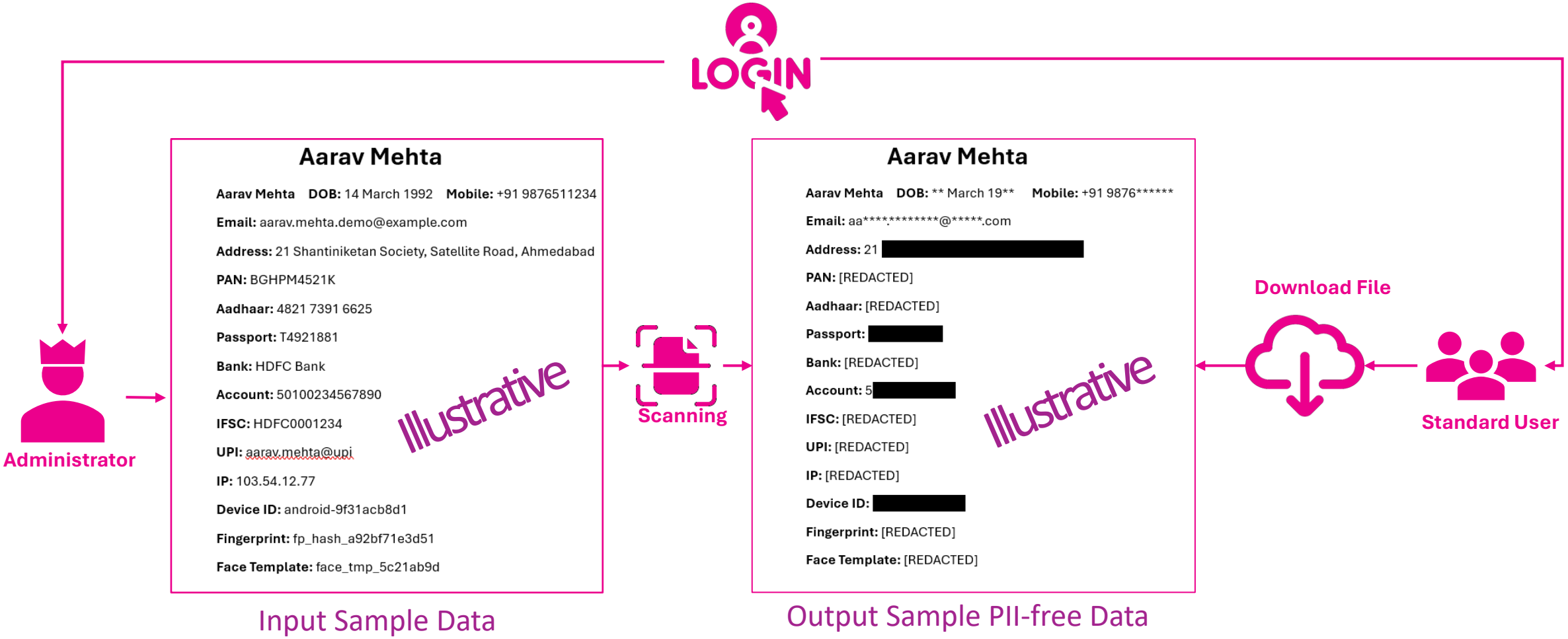
10. Sample Input-Output Visual



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10. Sample Input-Output Visual

DOCX / PDF

Michael B. Johnson

PERSONAL IDENTIFIABLE INFORMATION PROFILE

Full Name: Michael Benjamin Johnson
SSN: 321-45-7890
DOB: May 12, 1985
Current Address: 456 Liberty Ave, Unit 2B, Springfield, IL 61111
Home Phone: (217) 555-0123
Mobile Phone: (217) 555-8845
Personal Email: mb.johnson@emailprovider.com

SAMPLE DATA

EMPLOYMENT & FINANCIAL INFO

Emp ID: M-7741 **Acct:** 5432109876
Route: 7741-2233 **Route:** 123-45-6789

Michael B. Johnson

PERSONAL IDENTIFIABLE INFORMATION PROFILE

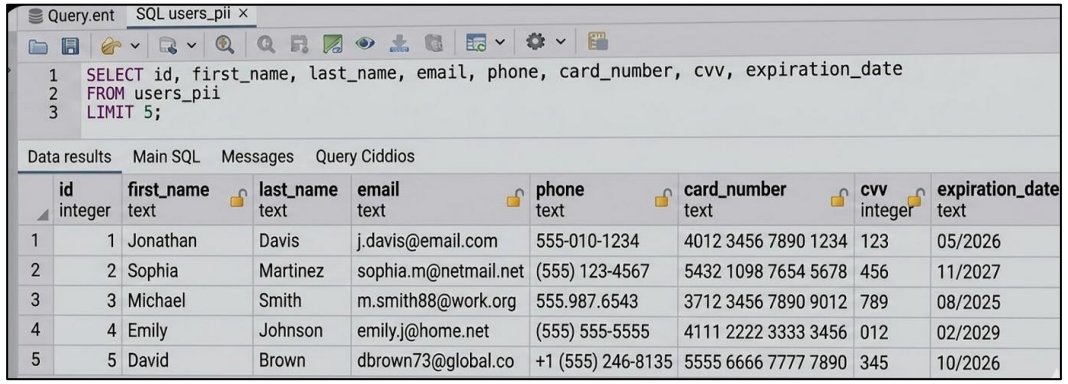
Full Name: Michael Benjamin Johnson
SSN: XXX-67-XXXX
DOB: May 12, 19XX
Current Address: 456 Liberty Ave, Unit 2B, Springfield, IL 62704
Home Phone: (217) 555-01XX
Mobile Phone: (217) 555-88XX
Personal Email: mb.johnson@emailprovider.com

SAMPLE DATA

EMPLOYMENT & FINANCIAL INFO

Emp ID: M-77XX **Acct:** XXXX-XXXX-3344
Route: XXXX-3344 **Route:** XXX-00-1111

SQL

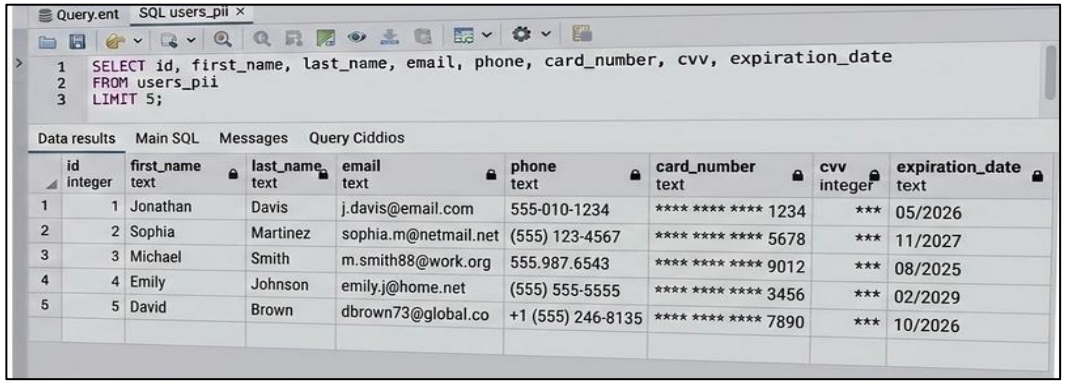


Query.ent | SQL users_pii

```
SELECT id, first_name, last_name, email, phone, card_number, cvv, expiration_date
FROM users_pii
LIMIT 5;
```

Data results | Main SQL | Messages | Query Ciddios

	id integer	first_name text	last_name text	email text	phone text	card_number text	cvv integer	expiration_date text
1	1	Jonathan	Davis	j.davis@email.com	555-010-1234	4012 3456 7890 1234	123	05/2026
2	2	Sophia	Martinez	sophia.m@netmail.net	(555) 123-4567	5432 1098 7654 5678	456	11/2027
3	3	Michael	Smith	m.smith88@work.org	555.987.6543	3712 3456 7890 9012	789	08/2025
4	4	Emily	Johnson	emily.j@home.net	(555) 555-5555	4111 2222 3333 3456	012	02/2029
5	5	David	Brown	dbrown73@global.co	+1 (555) 246-8135	5555 6666 7777 7890	345	10/2026



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Data results | Main SQL | Messages | Query Ciddios

	id integer	first_name text	last_name text	email text	phone text	card_number text	cvv integer	expiration_date text
1	1	Jonathan	Davis	j.davis@email.com	555-010-1234	**** **** **** 1234	***	05/2026
2	2	Sophia	Martinez	sophia.m@netmail.net	(555) 123-4567	**** **** **** 5678	***	11/2027
3	3	Michael	Smith	m.smith88@work.org	555.987.6543	**** **** **** 9012	***	08/2025
4	4	Emily	Johnson	emily.j@home.net	(555) 555-5555	**** **** **** 3456	***	02/2029
5	5	David	Brown	dbrown73@global.co	+1 (555) 246-8135	**** **** **** 7890	***	10/2026

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11. Selection Criteria (Evaluation)

File Processing Requirement

- Demonstrate PII masking capability
- at least 3 sample files

Mandatory File Formats

SQL PDF DOC

Mandatory formats must be included:
- SQL file
- PDF file
- DOCX file

The remaining files can be any format (e.g., TXT, CSV, JSON, etc.).

Input Output Demo

IN OUT

- Original Input File
- Processed Output File (with masked PII)

✓ PII fields masked
✓ other data remains intact

Implementation Format

- script-based solution (CLI)
- GUI-based tool (preferred)

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12. Download the Sample Dataset

Scan Me

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